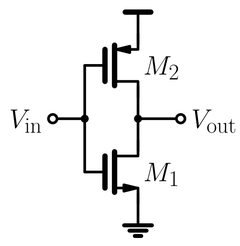
JOHANNES KEPLER UNIVERSITY LINZ

Fifty+ Nifty Variations of Two-Transistor Circuits

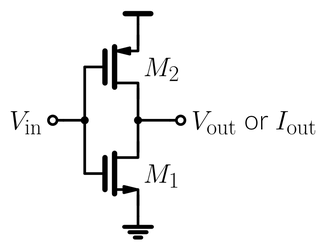
INSTITUTE FOR INTEGRATED CIRCUITS

Circuit No. 1



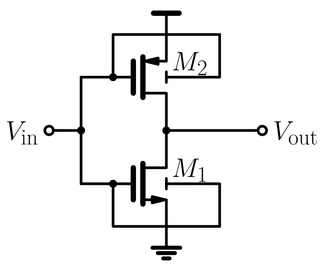
This is the ubiquitous digital **inverter**. The input voltage V_{in} switches one of both transistors on while the other is off.

Circuit No. 2



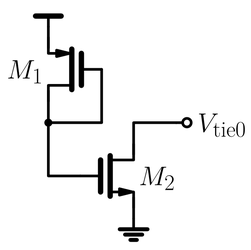
The same structure can be used as a **voltage amplifier** (V_{out}, with high-Z loading) or low-voltage complementary **transconductance stage** (I_{out}, with low-Z loading) when both MOS-FET are held in saturation.

Circuit No. 3



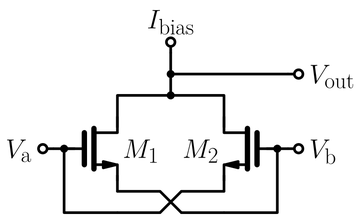
The (dynamic threshold) **DT-MOS inverter** achieves an improved current drive at low leakage current. It needs to be operated at low supply voltages to avoid a forward bias of the well diodes.

Circuit No. 4



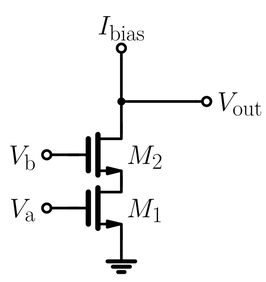
An ESD-safe **tie-zero** for unused CMOS logic inputs (no MOS-FET gate should be tied directly to a supply rail). The tie-one can be constructed accordingly.

Circuit No. 5



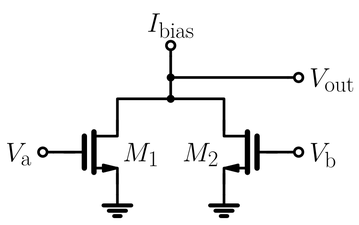
Using a current source I_{bias} with finite output impedance to bias this structure, this circuit implements an **XNOR** logic function ($V_{out} = \overline{V_a \oplus V_b}$). The logic inputs V_a and V_b must be driven by low-ohmic logic levels between V_{DD} and V_{SS}.

Circuit No. 6



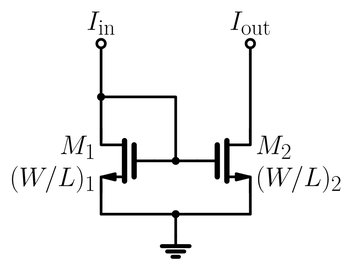
This series connection of two MOS-FETs realizes a logical **NAND** function ($V_{out} = \overline{V_a \wedge V_b}$) when driven by logic inputs.

Circuit No. 7



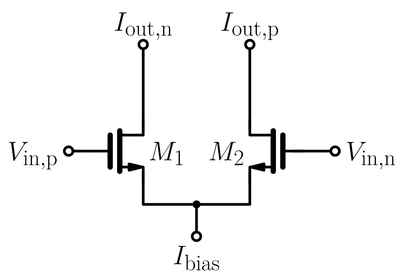
This circuit complements the logic gates and realizes a **NOR** function ($V_{out} = \overline{V_a \vee V_b}$).

Circuit No. 8



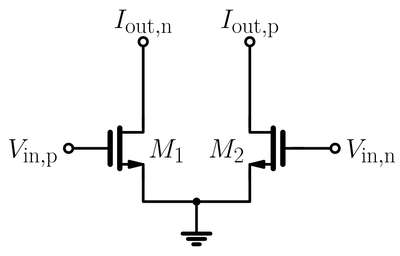
This circuit is the basic **current mirror**, simultaneously copying and sizing of $I_{out} = (W/L)_2/(W/L)_1 \cdot I_{in}$ according to the dimensions of M_1 and M_2.

Circuit No. 9



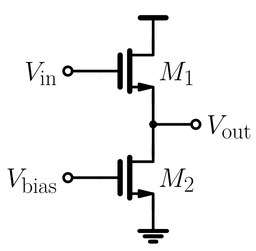
The ubiquitous **differential pair** (or long-tailed pair), like the current mirror, is a fundamental building block in integrated circuits.

Circuit No. 10



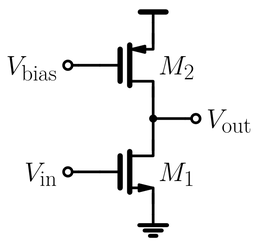
The **pseudo-differential pair** spares the tail current source's headroom in exchange for reduced common-mode rejection, but with the benefit of class-AB operation.

Circuit No. 11



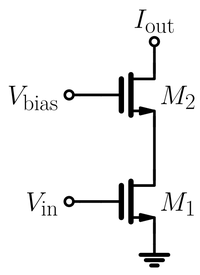
This is the **source follower** (or common-drain stage), utilizing M_2 as a current source to bias M_1.

Circuit No. 12



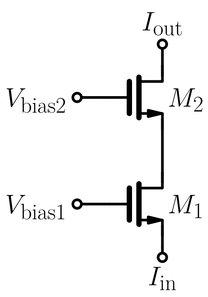
The **common-source** amplifier stage with active load.

Circuit No. 13



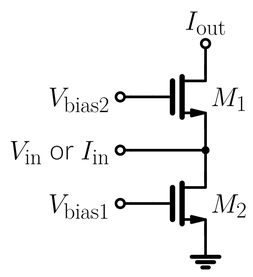
The **cascoded common-source** stage boosts the output impedance of M_1 considerably to $r_{out} \approx g_{m2}/(g_{ds1} \cdot g_{ds2})$.

Circuit No. 14



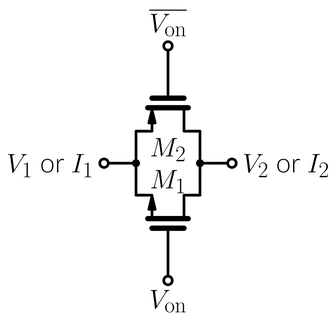
This is the **cascoded common-gate** stage. Note that $I_{out} \approx I_{in}$, but the impedance level changes drastically, creating gain or a high output impedance at the output node.

Circuit No. 15



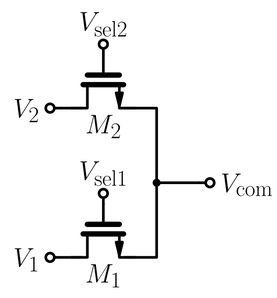
The **common-gate stage**, employing M_2 as a current source to bias transistor M_1. The input can be either a voltage- or a current-signal.

Circuit No. 16



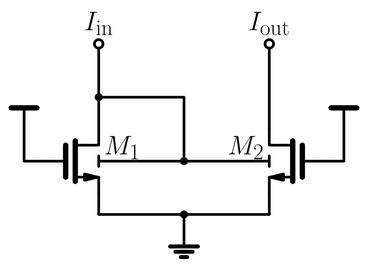
The **transmission gate** switches either voltage (V_1 and V_2) or current (I_1 and I_2) (and it works rail to rail, too).

Circuit No. 17



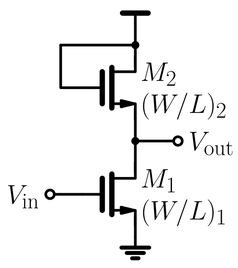
The 2-to-1 **multiplexer**, connecting either V_1 or V_2 to V_{com}. Depending on V_{sel1} and V_{sel2}, the MOS-FETs are alternately switched on or off.

Circuit No. 18



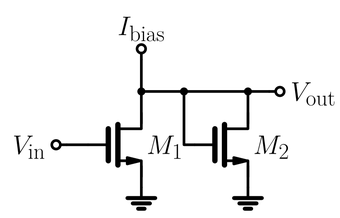
This circuit is an improved **bulk-driven current mirror** that allows low voltage operation, requiring a voltage headroom substantially less than V_{GS1}.

Circuit No. 19



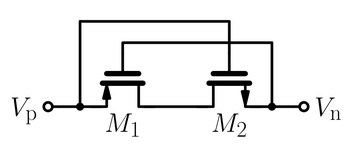
The **common-source amplifier with diode load** is sometimes called a wide-band amplifier due to its potentially high-speed operation. Here, the gain is set precisely to $A_v = V_{out}/V_{in} = -\sqrt{(W/L)_1/(W/L)_2}$, only depending on transistor sizing (if we neglect the body effect and finite output conductance).

Circuit No. 20



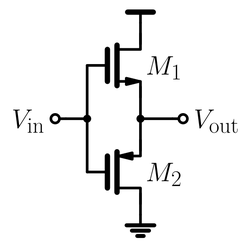
This **wide-band amplifier** has the advantage of a removed body effect in M_2 and a ground-referred output node.

Circuit No. 21



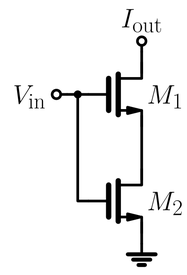
This **ultra-low-power diode (ULPD)** shows a reduced leakage in the reverse direction when $V_n > V_p$.

Circuit No. 22



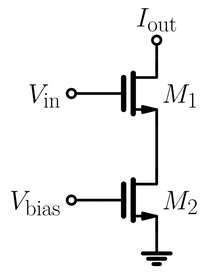
This class-B **push-pull follower** can be considered an enhanced version of the simple source follower. However, lacking a class-A bias component, this structure is subject to cross-over distortion.

Circuit No. 23



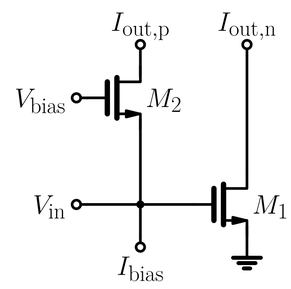
This circuit is a **MOS-FET-R degenerated common-source** stage. By sizing M_2 appropriately, the degeneration can be adapted. This arrangement using two transistors can also increase the length of a (compound) device, for example, in current mirrors, as otherwise, MOS-FETs with different L will not match well.

Circuit No. 24



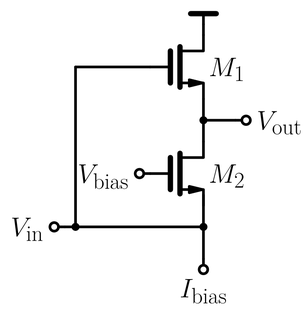
This is a variation of the implementation shown earlier, where the **degeneration** of M_1 can be **adapted** by tuning V_{bias}.

Circuit No. 25



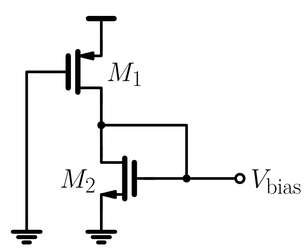
This **common-gate–common-source** topology offers an impedance-matched single-ended input and a differential output while simultaneously canceling noise and distortion.

Circuit No. 26



This **low-noise amplifier** was discovered by doing an exhaustive search for potential two-transistor wide-band amplifiers. For practical implementation, M_1 requires an ac-coupling (and proper biasing) in its gate connection to keep M_2 in saturation.

Circuit No. 27



This (simple) **bias voltage generator** uses the current source M_1 to bias M_2 so that $V_{bias} = V_{GS2}$. Note that the generated voltage is susceptible to changes in process, supply voltage, and temperature (PVT).

Based on: Harald Pretl and Matthias Eberlein, "Fifty Nifty Variations of Two-Transistor Circuits: A tribute to the versatility of MOSFETs," in *IEEE Solid-State Circuits Magazine*, Summer 2021.
doi: 10.1109/MSSC.2021.3088968, harald.pretl@jku.at

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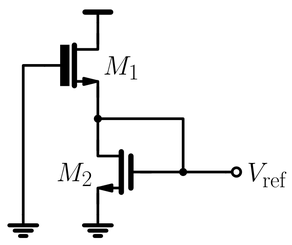
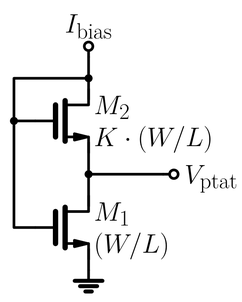
Circuit No. 28



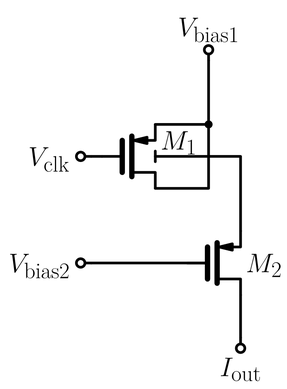
This **constant bias voltage** generator creates a remarkably stable output voltage (note that M_1 and M_2 must have different threshold voltages $V_{\mathrm{th1}} \neq V_{\mathrm{th2}}$).

Circuit No. 29



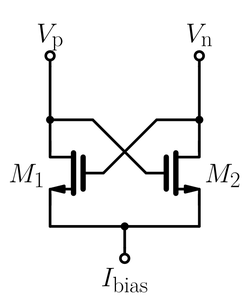
This circuit is a **PTAT voltage** generator, if M_1 and M_2 are kept in subthreshold operation.

Circuit No. 30



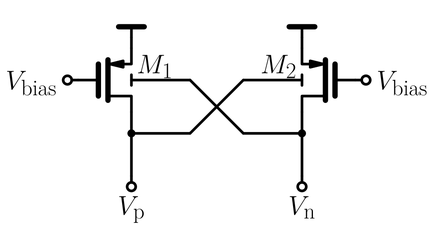
This **pA current source** is based on the periodic filling and flushing of the Si–SiO_2 interface traps by alternating M_1 between accumulation and inversion. It can operate with reasonably high clock frequencies and still create tiny currents.

Circuit No. 31



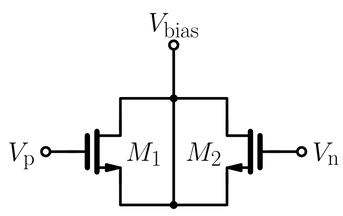
Two transistors with cross-coupling form a **negative resistance** between V_{p} and V_{n}, mainly employed in oscillators and comparators. As in a differential pair, the bias current source can be replaced by a fixed potential.

Circuit No. 32



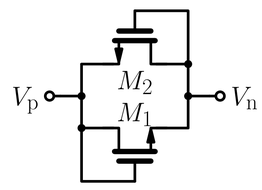
This circuit is a **low-voltage** version of a **cross-coupled pair**, where the body controls the MOS-FETs, avoiding the significant V_{GS} drop at V_{p} and V_{n}.

Circuit No. 33



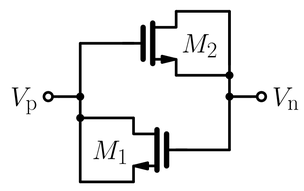
This **varactor** (the capacitance between V_{p} and V_{n} depends on the bias voltage V_{bias}) is often used in voltage-controlled oscillators. In most technologies, the NMOS can be put inside the n-well so that the varactor works in accumulation, providing an optimized tuning range and high Q.

Circuit No. 34



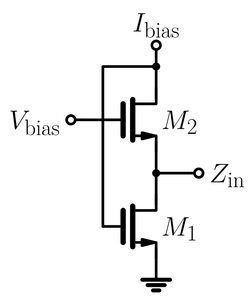
The **anti-parallel MOS-FET diodes** can be employed for many things, for example, voltage clamping.

Circuit No. 35



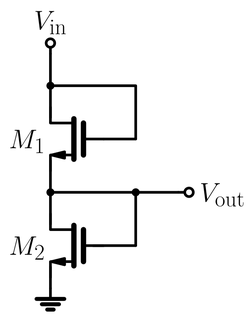
The **anti-parallel MOS-FET capacitors** make the differential capacitance between V_{p} and V_{n} more linear and symmetrical. As in a varactor, an NMOS–in–n-well is an option.

Circuit No. 36



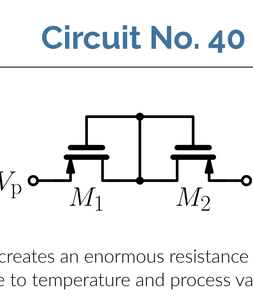
This circuit, which is a similar configuration as the flipped voltage follower, can function as an **active inductor**, providing $L = C_{\mathrm{GS1}}/(g_{m1} \cdot g_{m2})$.

Circuit No. 37



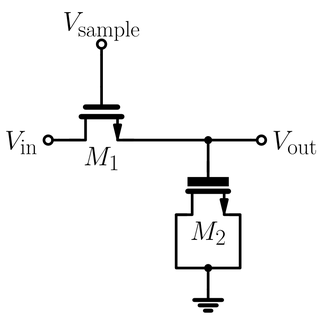
This circuit is an area-efficient **voltage divider**. If M_1 and M_2 are of the same size, then $V_{\mathrm{out}} \approx V_{\mathrm{in}}/2$. Often, a PMOS version is a better choice since it can avoid the body effect by tying the body to the respective source for M_1 and M_2.

Circuit No. 38



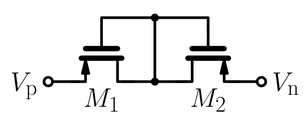
This implementation is the **Bult current divider** (if M_1 and M_2 are of identical size, then I_{in} is precisely split in half between I_{out1} and I_{out2}).

Circuit No. 39



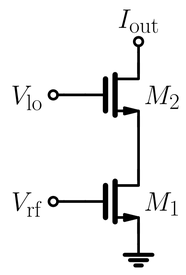
This is a **sample-and-hold** circuit as used in a lot of ADC implementations, using the gate capacitance of M_2 as a storage capacitor (a low to zero V_{th} would be an advantage in this case).

Circuit No. 40



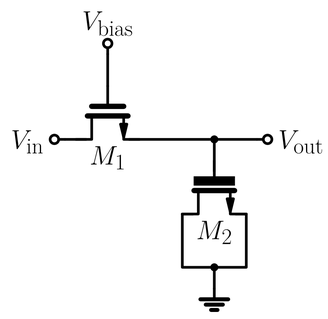
This arrangement creates an enormous resistance between V_{p} and V_{n}, although susceptible to temperature and process variations.

Circuit No. 41



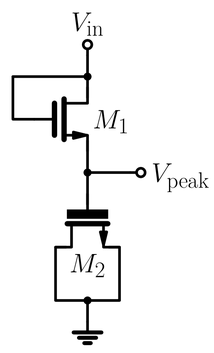
In this **dual-gate MOS-FET mixer**, the periodic local-oscillator signal V_{lo} causes the time-variant change of the transconductance of M_1, resulting in a frequency conversion from the input V_{rf} to the output I_{out}.

Circuit No. 42



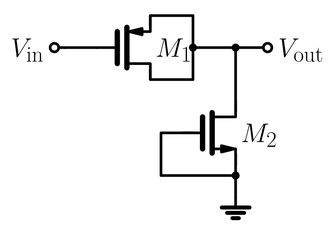
The sample-and-hold circuit becomes a continuous-time **low-pass filter** if M_1 is connected to a fixed bias voltage instead of a clock signal. Note that this circuit transforms into a high-pass filter when M_1 and M_2 are swapped.

Circuit No. 43



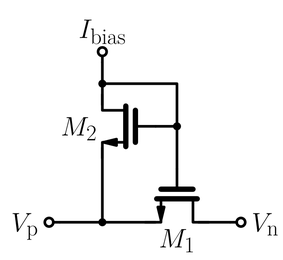
This circuit is a minimalistic **peak-voltage detector** (essentially a max-hold), where $V_{\mathrm{peak}} = V_{\mathrm{in,max}} - V_{\mathrm{gs1}}$.

Circuit No. 44



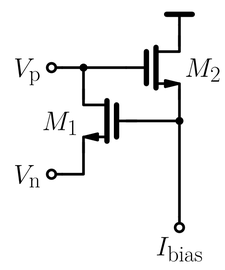
A similar circuit to the peak detector can function as an approximate **voltage doubler** when driven by a sinusoidal input voltage (on negative swings of V_{in} the capacitor M_1 gets charged to $|V_{\mathrm{in}}| - V_{\mathrm{gs2}}$, which is added to V_{in} during positive swings when M_2 is off). As in any circuit with negative voltages, proper connection of the wells is required.

Circuit No. 45



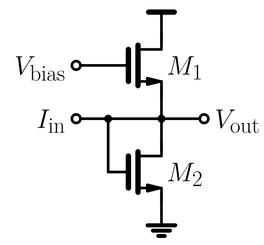
This circuit implements a current-controlled (high-impedance) floating **resistor** between V_{p} and V_{n}.

Circuit No. 46



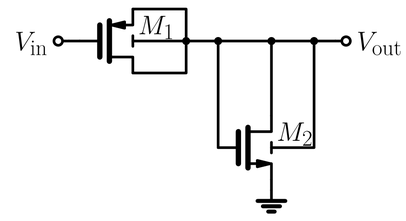
The floating **level shift** (or "floating **battery**") effectively shifts a bias point between V_{p} and V_{n}, as $V_{\mathrm{shift}} = V_{\mathrm{p}} - V_{\mathrm{n}} = V_{\mathrm{GS1}} + V_{\mathrm{GS2}}$.

Circuit No. 47



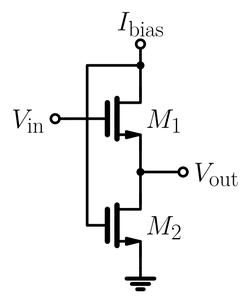
This circuit is a perfectly **linear I-to-V converter** with $V_{\mathrm{out}}/I_{\mathrm{in}} = [\mu C_{\mathrm{ox}} \cdot (W/L) \cdot (V_{\mathrm{bias}} - 2V_{\mathrm{th}})]^{-1}$, if we assume a square-law behavior, neglect the body effect, and M_1 and M_2 are of same size and kept in saturation.

Circuit No. 48



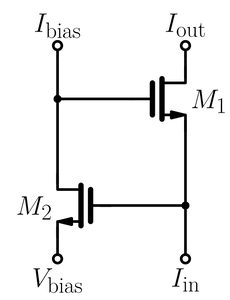
This **level-shifter circuit** shifts a digital input voltage V_{in} to an output voltage V_{out} swinging around V_{SS}.

Circuit No. 49



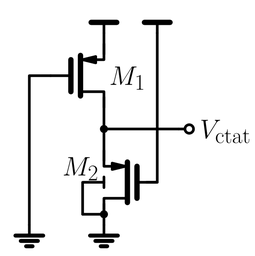
The **flipped voltage follower** is an improved version of the standard source follower, employing feedback to lower the output impedance to $r_{\mathrm{out}} = g_{ds2}/(g_{m1} \cdot g_{m2})$.

Circuit No. 50



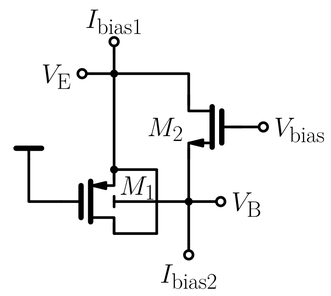
The **regulated cascode** improves the effect of the cascode M_1 by g_{m2}/g_{ds2} due to feedback. Note that the source of M_2 can be tied to ground if combined with a common-source input stage.

Circuit No. 51



Exploiting the parasitic (lateral) PNP transistor inherent in a PMOS structure, this simple **CTAT voltage generator** can be created.

Circuit No. 52



The parasitic BJT (lateral or vertical) often suffers from poor $\beta \ll 10$. This circuit forces the collector current of the parasitic (vertical) **PNP** (realized by M_1) to $I_C = I_{\mathrm{bias1}} - I_{\mathrm{bias2}}$, although the collector terminal (being the p-substrate) is not accessible. By doing this, the resulting $V_{\mathrm{EB}} = V_{\mathrm{E}} - V_{\mathrm{B}}$ can be accurately used in a bandgap circuit.

Based on: Harald Pretl and Matthias Eberlein, "Fifty Nifty Variations of Two-Transistor Circuits: A tribute to the versatility of MOSFETs," in *IEEE Solid-State Circuits Magazine*, Summer 2021.
doi: 10.1109/MSSC.2021.3088968, harald.pretl@jku.at